Benchmarking Robustness of Locomotion

Thesis: Robustness is equivalent to responsiveness of robot

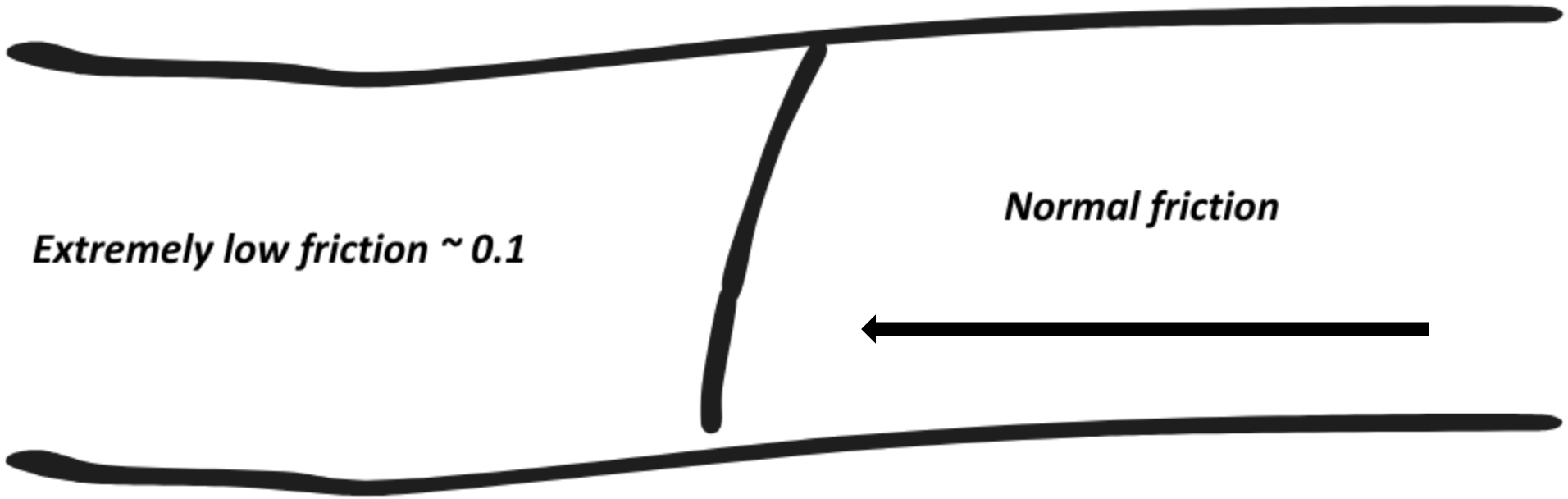
***Key Challenge**: Partial information, field of view and obscurance*

Required Features:

- *Feet reflex*
- *Affordance estimation (implicit or explicit)*
- *Trajectory correction*

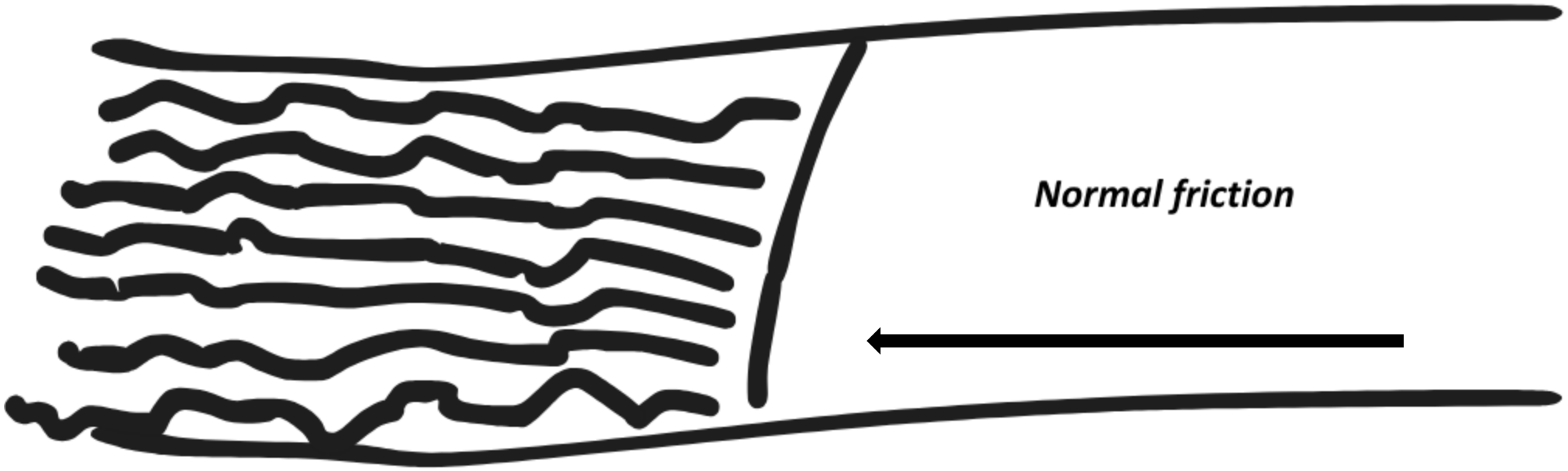
Benchmark should test feet reflex and trajectory correction across a canonical set of scenarios necessarily out of distribution

Feet Reflex: Scenario 1 (RS1): Extremely Low Friction Transition



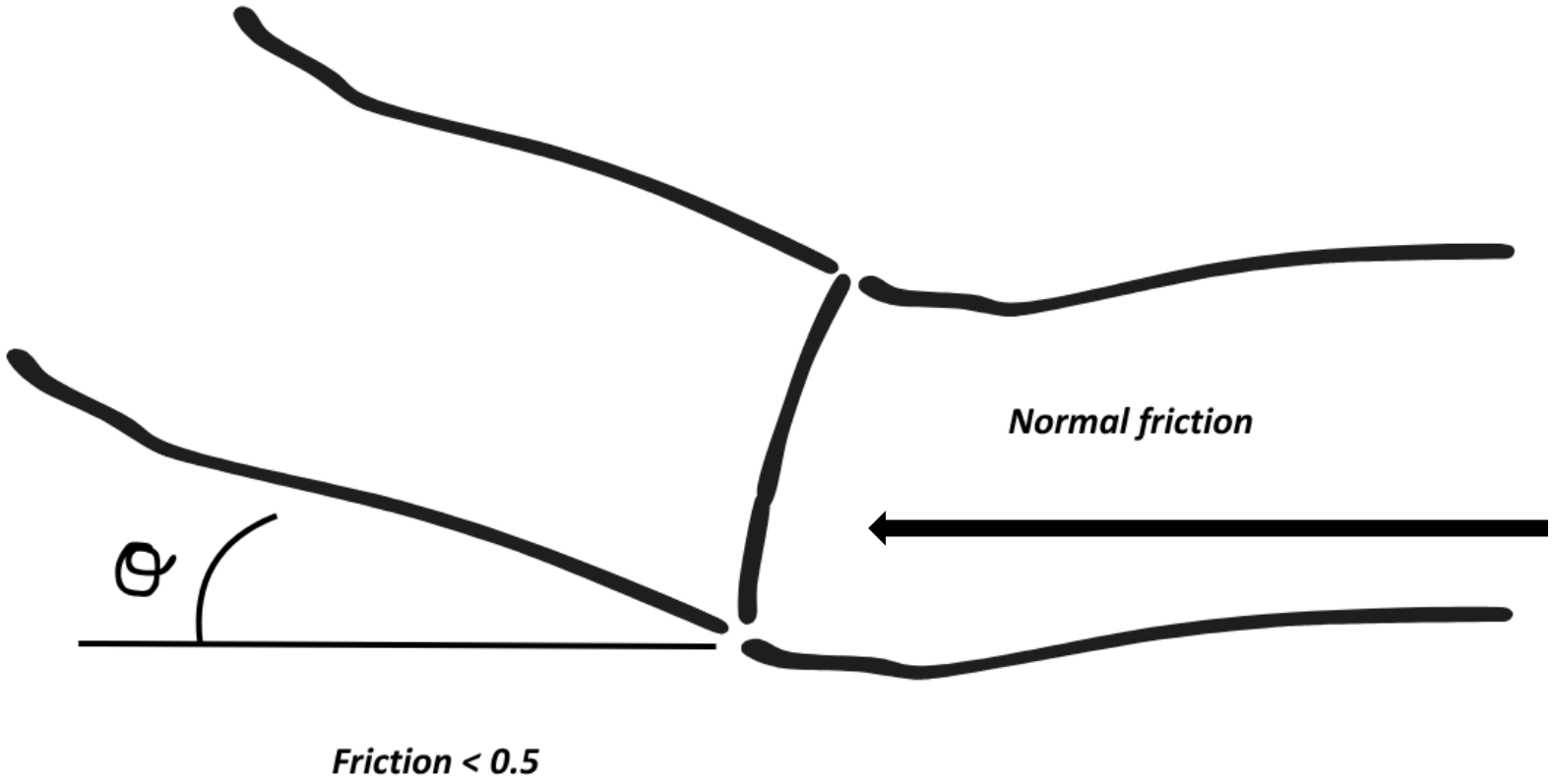
Can the robot sustain its motion upon sudden change of friction? Upto what speed?

Feet Reflex: Scenario 2 (RS2): Extremely Low Friction on Rough



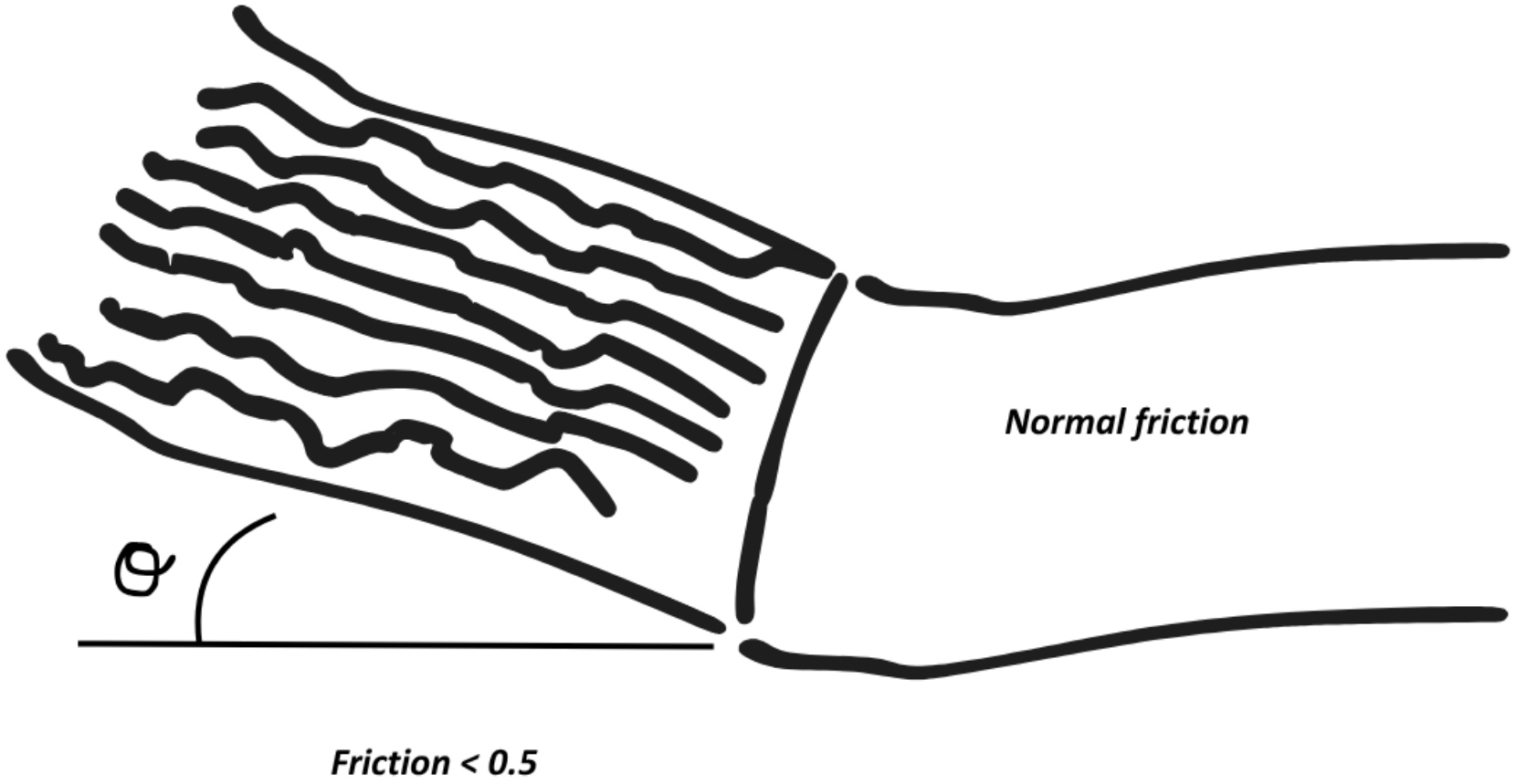
Can the robot sustain its motion upon sudden change of friction and roughness? Upto what speed? Upto what roughness

Feet Reflex: Scenario 3 (RS3): Low Friction on Slope



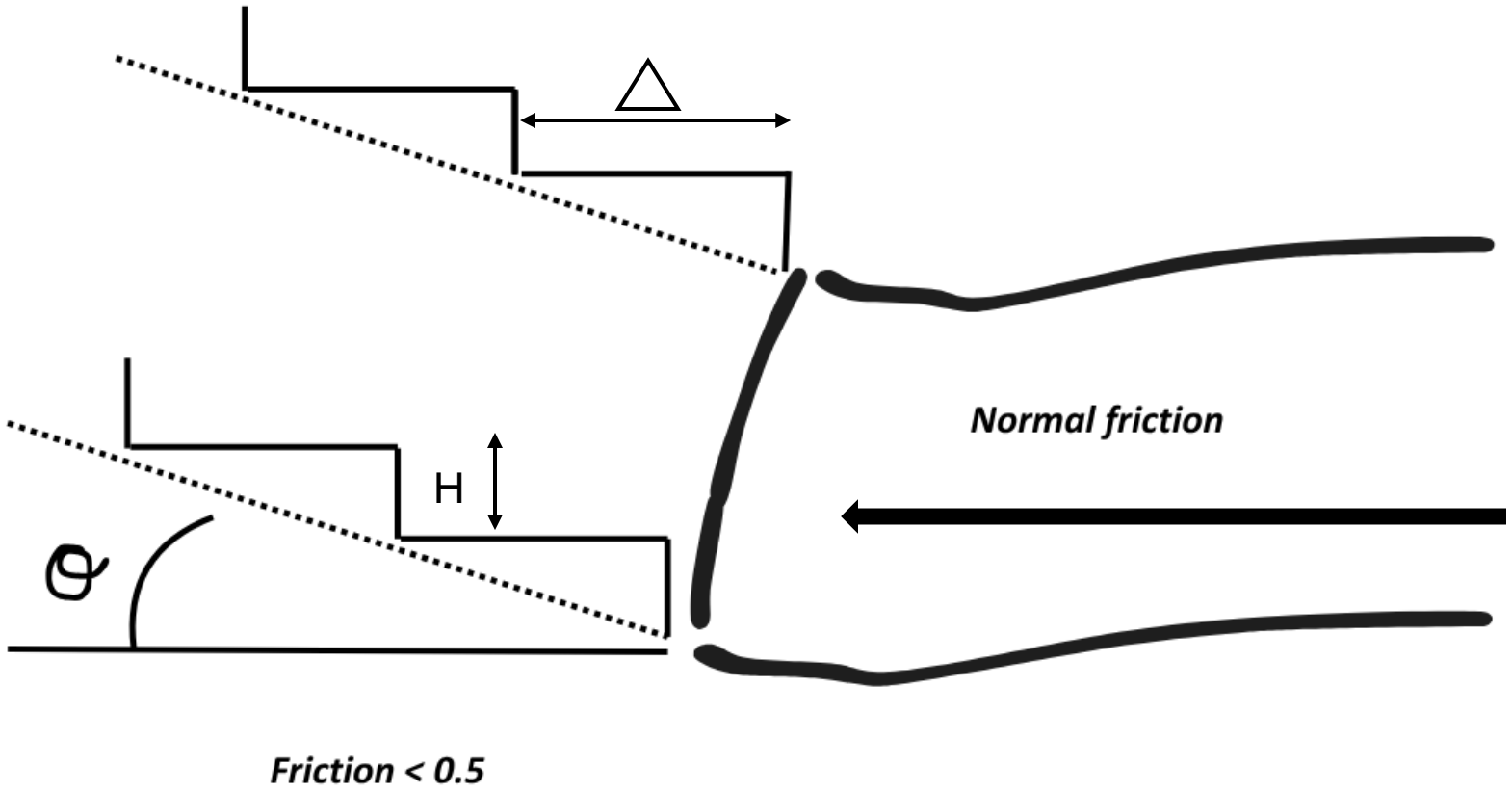
Can the robot sustain its motion upon sudden change of friction and slope? Upto what speed? Upto what angle?

Feet Reflex: Scenario 4 (RS4): Low Friction on Rough Slope



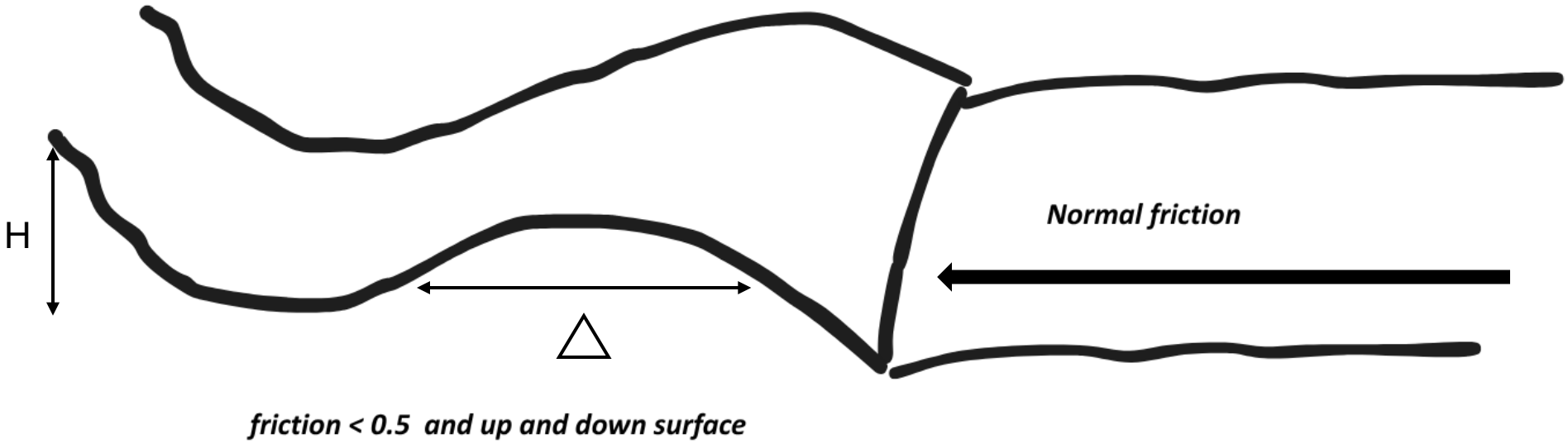
Can the robot sustain its motion upon sudden change of friction and slope where the slope is rough? Upto what speed? Upto what angle

Feet Reflex: Scenario 5 (RS5): Low Friction on Staircase



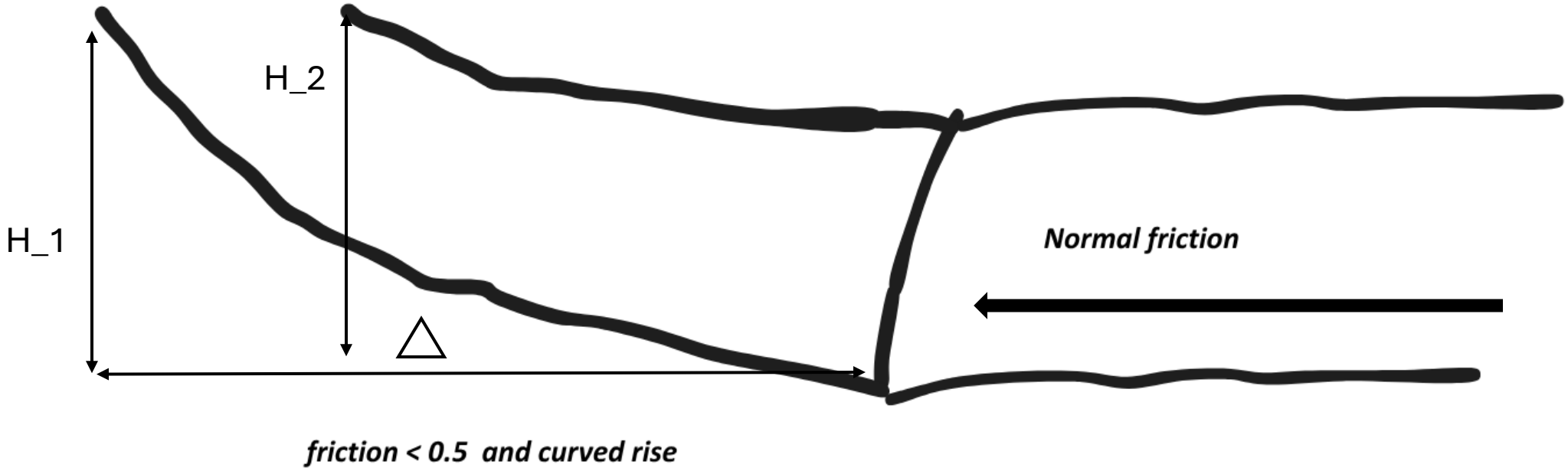
Can the robot sustain its motion upon sudden change of friction and slope where the slope is terraced? Additionaly the surface can be rough. Upto what speed? Upto what angle

Feet Reflex: Scenario 6 (RS6): Low Friction on Up and Down



Can the robot sustain its motion upon sudden change of friction and up and down surface nature? Upto what speed? Upto what minimal △ and maximum H? Note: the limit of this is the rough scenario.

Feet Reflex: Scenario 7 (RS7): Low Friction on Curved Slope



Can the robot sustain its motion upon sudden change of friction and a curved slope? Upto what speed? Upto what difference H_1 – H_2?

Feet Reflex: Scenario 8 (RS8): Repeated small gaps

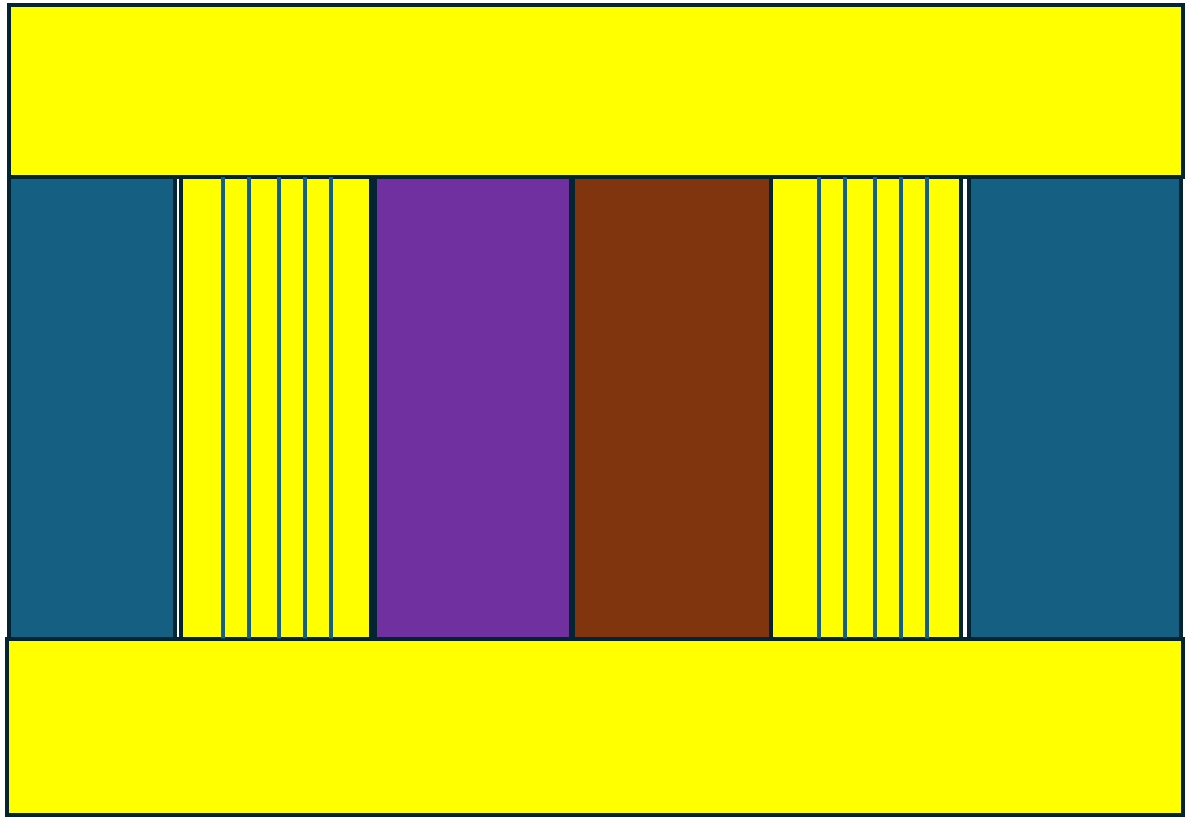
Feet Reflex: Scenario 9 (RS9): Large gaps

Feet Reflex: Scenario 10 (RS10): High Rise

Feet Reflex: Scenario 10 (RS11): Swamp and Marsh

Affordance Estimation

Affordance Estimation: Scenario (AE): Jump/Climb or Not?



Color Code:

- Nominal surface
- Surface with Friction and Roughness
- Surface with Friction and Roughness and RS8
- RS1 or RS2 or RS3 or RS4 or RS5 or RS6 or RS7 or RS9 or RS10 or RS 11
- RS1 or RS2 or RS3 or RS4 or RS5 or RS6 or RS7 or RS9 or RS10 or RS 11

Affordance Estimation: Scenario (AEE)

